

One of the most useful results is the LT of $\frac{df}{dt}$, which we have seen in the table, and which we now derive.

Example 6.4.2

have to think cleverly, use a few tricks when transforming

Show that

$$\mathcal{L}\left\{\frac{df}{dt}H(t)\right\} = sF(s) - f(0),$$

where $\mathcal{L}\{f(t)H(t)\} = F(s)$.**Solution.**

$$\begin{aligned}\mathcal{L}\left\{\frac{df}{dt}H(t)\right\} &= \int_0^{\infty} \frac{df}{dt} H(t) e^{-st} dt \quad \text{in integral range} \\ &= [e^{-st} f(t)]_0^{\infty} - \int_0^{\infty} f(t) (-se^{-st} dt) \\ &= \left(\lim_{t \rightarrow \infty} e^{-st} f(t) - e^0 f(0)\right) + s \int_0^{\infty} f(t) e^{-st} dt \\ &= \lim_{t \rightarrow \infty} e^{-st} f(t) - f(0) + sF(s) \quad \text{Laplace transform definition}\end{aligned}$$

We don't really need a new variable t

From p6, if $|f(t)| < Me^{ct}$ then $F(s)$ exists for $s > c$
 $\Rightarrow |e^{-st} f(t)| < Me^{-st} e^{ct} = Me^{(c-s)t} \rightarrow 0$ as $t \rightarrow \infty$
 (neg. since $s > c$)

$$\begin{aligned}&= \lim_{t \rightarrow \infty} e^{-st} f(t) - f(0) + sF(s) \\ &= sF(s) - f(0) \leftarrow \text{algebraic! (no derivatives)}\end{aligned}$$

use this definition

The definition and for the Laplace Transform of a function $f(t)$ when $t \geq 0$ is:

$$F(s) = \int_0^{\infty} f(t) e^{-st} dt = \mathcal{L}\{f(t)\}.$$

integration by parts:
 $\int u dv = uv - \int v du$
 Let $u = e^{-st}$
 $du = -se^{-st} dt$
 $dv = \frac{df}{dt} dt$
 $v = f(t)$



Note that when we transform the derivative of a function we also have to supply an initial value – $f(0)$ – of that function. The final piece of the puzzle which will transform ODEs into algebraic equations is to prove the *linearity* of Laplace transforms.

6.5 LINEARITY

A very important property of the Laplace Transform is *linearity*, which is what lets us transform equations and formulas term by term. If a, b are constants and $f(t), g(t)$ are functions, then

$$\begin{aligned}\mathcal{L}\{af(t) + bg(t)\} &= \int_0^\infty (af(t) + bg(t))e^{-st} dt \\ &= a \int_0^\infty f(t)e^{-st} dt + b \int_0^\infty g(t)e^{-st} dt\end{aligned}$$

$$\mathcal{L}\{af(t) + bg(t)\} = a\mathcal{L}\{f(t)\} + b\mathcal{L}\{g(t)\}.$$

very useful

With the linearity rule, formulas with added or subtracted terms can be transformed term-by-term.

└ not multiplied!

6.6 MORE LAPLACE TRANSFORMS

LAPLACE TRANSFORMS OF POLYNOMIALS

Example 6.6.1

If $n > 0$ is an integer, show that

$$\mathcal{L}\{t^n H(t)\} = \frac{n!}{s^{n+1}}.$$

Solution.



This result, combined with linearity, enables us to find the LT of any polynomial.

LAPLACE TRANSFORMS OF TRIG FUNCTIONS**Example 6.6.2**

Show that

$$\mathcal{L}\{\sin(\omega t)H(t)\} = \frac{\omega}{s^2 + \omega^2},$$

and hence show that

$$\mathcal{L}\{\cos(\omega t)H(t)\} = \frac{s}{s^2 + \omega^2}.$$

Solution.



LAPLACE TRANSFORMS OF HIGHER DERIVATIVES

We find $\mathcal{L}\left\{\frac{d^2f}{dt^2}H(t)\right\}$ and the LT of higher derivatives by an iterative process.

$$\begin{aligned}\mathcal{L}\left\{\frac{d^2f}{dt^2}H(t)\right\} &= \mathcal{L}\left\{\frac{d}{dt}\left(\frac{df}{dt}\right)H(t)\right\} = s\mathcal{L}\left\{\frac{df}{dt}H(t)\right\} - \frac{df}{dt}\bigg|_{t=0} \\ &= s(s\mathcal{L}\{f(t)H(t)\} - f(0)) - f'(0)\end{aligned}$$

$$\mathcal{L}\left\{\frac{d^2f}{dt^2}H(t)\right\} = s^2F(s) - sf(0) - f'(0).$$

In general:

$$\mathcal{L}\{f^{(n)}(t)\} = s^nF(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \dots - f^{(n-1)}(0).$$

We can now tackle some harder problems, such as the following second order ODE.

Example 6.6.3

Solve the initial value problem

$$y'' - y' - 6y = 2, \quad y(0) = 1, \quad y'(0) = 0.$$

Solution.



Recall once more that we have already inserted the initial conditions: this solution satisfies the ODE *and* the initial conditions.

LAPLACE TRANSFORM OF AN INTEGRAL

From time to time, we may wish to take the LT of an integral. We can derive this in a roundabout but clever way, as follows. Suppose that

$$f(t) = \int_{\tau=0}^t g(\tau) d\tau$$

is the integral whose LT we wish to find. The fundamental theorem of calculus² then tells us that

$$\frac{df}{dt} = g(t)$$

while the definition of $f(t)$ shows us that $f(0) = 0$. We can put all of this together with our knowledge that

$$\mathcal{L}\left\{\frac{df}{dt}H(t)\right\} = s\mathcal{L}\{f(t)\} - f(0)$$

to get

$$\mathcal{L}\{g(t)H(t)\} = s\mathcal{L}\left\{\int_{\tau=0}^t g(\tau) d\tau\right\} - 0,$$

which yields

$$\mathcal{L}\left\{\int_{\tau=0}^t g(\tau) d\tau\right\} = \frac{1}{s}G(s),$$

where $G(s) = \mathcal{L}\{g(t)\}$.

²Do you need to look this up?

6.7 FIRST SHIFT THEOREM

This fact will enable us to tackle a wider range of problems.

Example 6.7.1

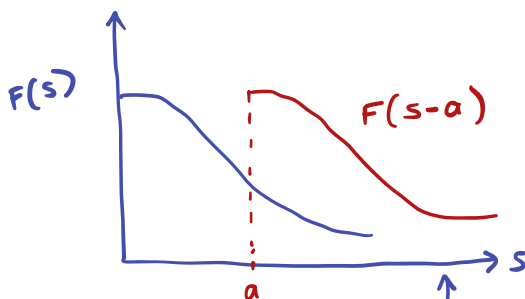
$$\mathcal{L}\{f(t)\} = F(s)$$

Prove the *First Shift Theorem*, namely that

$$\mathcal{L}\{e^{at}f(t)H(t)\} = F(s-a).$$

Solution.

$$\begin{aligned}\mathcal{L}\{e^{at}f(t)H(t)\} &= \int_0^\infty e^{at}f(t)e^{-st}dt \\ &= \int_0^\infty f(t)e^{(a-s)t}dt \\ &= \int_0^\infty f(t)e^{-(s-a)t}dt \\ &= \int_0^\infty f(t)e^{-\sigma t}dt \quad \text{Let } \sigma = s-a \\ &= F(\sigma) \\ &= F(s-a)\end{aligned}$$



This result is called the first *shift* theorem because it says that the LT of $e^{at}f(t)$ is the LT of $f(t)$ shifted by a to the right. For example, earlier we saw that

$$\mathcal{L}\{\underbrace{\sin(\omega t)}_{f(t)}H(t)\} = \frac{\omega}{s^2 + \omega^2} = F(s)$$

The first shift theorem tells us that

$$\mathcal{L}\{e^{at}\sin(\omega t)H(t)\} = \frac{\omega}{(s-a)^2 + \omega^2} = F(s-a) \quad (1)$$

We also saw that

$$\mathcal{L}\{\underbrace{\cos(\omega t)}_{f(t)}H(t)\} = \frac{s}{s^2 + \omega^2}$$

The only effect this has is to replace all s terms with $(s-a)$.

and so in a similar manner the first shift theorem tells us that

$$\mathcal{L}\{e^{at}\cos(\omega t)H(t)\} = \frac{s-a}{(s-a)^2 + \omega^2} = F(s-a) \quad (2)$$

The first shift theorem enables us to find the inverse LT of a wider variety of fractions which may result from running through the four steps of solving an ODE by LTs. Consider the following example.

Example 6.7.2

Find $y(t)$ when

$$Y(s) = \frac{s}{s^2 - 10s + 29}.$$

can't factorize nicely
 $\Rightarrow$ not partial fractions

Solution.

Complete the square:

$$s^2 - 10s + 29 = (s - 5)^2 - 25 + 29 = (s - 5)^2 + 4$$

$$\rightarrow Y(s) = \frac{s \text{ } \boxed{-5 + 5}}{(s - 5)^2 + 4} \quad \leftarrow \text{similar to } \mathcal{L}\{e^{5t} \cos 2t\} \text{ except numerator which needs to be } s - 5$$

$$= \frac{s - 5}{(s - 5)^2 + 2^2} + \frac{5}{(s - 5)^2 + 2^2} \times \frac{2}{2}$$

$$= \frac{s - 5}{(s - 5)^2 + 2^2} + \frac{5}{2} \frac{2}{(s - 5)^2 + 2^2}$$



When using the first shift theorem related to sine and cosine functions — results (1) and (2) — one has to be extremely careful about the different role played by s and ω .